Detecting Factual Hallucinations in Retrieval-Augmented Generation Systems:

A Comparative Study of Faithfulness Detection Methods

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Abstract

Retrieval-Augmented Generation (RAG) systems ground language model outputs in retrieved documents, but they do not eliminate the risk that generated text will contradict or misrepresent those documents. This paper evaluates multiple methods for detecting such faithfulness failures on the HaluEval QA benchmark (10,000 examples): zero-shot Natural Language Inference (NLI) using facebook/bart-large-mnli and MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli, a SummaC-Zero sentence-level aggregation variant, supervised fine-tuning of microsoft/deberta-v3-small, and a planned LLM-as-Judge baseline. On the HaluEval test set, both zero-shot NLI models achieved 68.8% accuracy and an F1-score of 0.657, while the SummaC-Zero variant scored 67.2% accuracy (F1 = 0.642). The fine-tuned DeBERTa-v3-small model failed to converge during training: across two data-split configurations (80/20 and 70/15/15), the model consistently predicted a single class for all examples, yielding approximately 50% accuracy and an artificially inflated F1 driven entirely by perfect recall on one class. Training loss remained near 0.693—the theoretical loss for random binary prediction—indicating that the model did not learn discriminative features from the HaluEval data under the chose hyperparameter configuration. These results demonstrate that zero-shot NLI provides a reasonable baseline for hallucination detection without any task-specific training, but that naive fine-tuning of a smaller model on this benchmark is not straightforward and requires careful hyperparameter tuning, data formatting, or architectural adjustments to succeed. The LLM-as-Judge evaluation remains ongoing as future work.

1. Introduction

Large language models (LLMs) are capable of generating fluent, grammatically correct text on virtually any topic, but this fluency does not guarantee factual accuracy. When these models are deployed without access to external knowledge, they rely entirely on parameters encoded during training, which can be outdated, incomplete, or internally inconsistent.Retrieval-Augmented Generation(RAG) addresses limitation by conditioning model generation on a set of documents retrieved at inference time. The model's output is therefore supposed to be grounded in evidence that the system can inspect and, in principle, verify.

Despite this grounding mechanism, RAG systems still produce outputs that deviate from the retrieved documents. The deviation may be intrinsic—where the model generates text that is internally inconsistent within the document set—or extrinsic, where the model introduces information that has no support in the retrieved passages. Both forms of deviation are broadly referred to as hallucinations. Detecting hallucinations in RAG outputs is a prerequisite for deploying these systems in settings where accuracy matters, such as question answering over legal, medical, or scientific corpora.

This study focuses on the faithfulness axis of hallucination: whether a model's generated answer is entailed by, contradicted by, or neutral with respect to the document from which it was purportedly derived. The project evaluates three automated detection strategies of varying complexity—NLI-based zero-shot inference, task-specific fine-tuning, and LLM-based judgment—and characterizes the trade-offs among them in terms of accuracy, F1-score, and cross-domain generalizability.

The remainder of this report is organized as follows. Section 2 reviews relevant prior work. Section 3 describes the experimental methods in detail. Section 4 presents quantitative results. Section 5 discusses the implications of those results, including an analysis of training failure modes. Section 6 offers conclusions, Section 7 provides acknowledgements, and Section 8 lists references.

2. Related Work

2.1 Retrieval-Augmented Generation

Lewis et al. (2020) introduced the RAG framework as a general-purpose architecture for knowledge-intensive NLP tasks. Their system combines a dense passage retriever—a bi-encoder trained with the DPR objective—with a sequence-to-sequence generator, specifically BART. At inference time, the retriever selects the top-k passages from a Wikipedia document index, and these passages are concatenated to the input query before being processed by the generator. The model is trained end-to-end, with gradients flowing through the generator but not the retriever's document index, keeping retrieval computationally tractable.

Lewis et al. reported state-of-the-art results on Open-Domain Question Answering benchmarks including Natural Questions, TriviaQA, and WebQuestions. Their analysis showed that the retriever learned to surface passages that directly supported correct answers, but they did not address the case where the generator's output diverges from the retrieved passage content. This gap—generating text that contradicts retrieved evidence—is precisely the problem this study targets.

The original RAG architecture has since been extended in numerous directions, including iterative retrieval, query reformulation, and passage reranking. However, faithfulness verification at the output stage remains a largely separate concern that the retrieval mechanism alone cannot address.

2.2 NLI-Based Consistency Detection

Laban et al. (2021) proposed a pipeline for detecting factual inconsistencies in abstractive summaries that translates naturally to the RAG faithfulness problem. Their method, published as SummaC, decomposes the generated summary into individual sentences and computes entailment scores between each summary sentence and each sentence in the source document using a pre-trained NLI model. The sentence-pair scores are then aggregated—either by taking the maximum score across source sentences or by using a learned aggregation model—to produce a document-level consistency score.

SummaC was evaluated on multiple summarization consistency benchmarks and demonstrated that NLI-based models, when applied at the sentence level rather than the document level, provide substantially stronger consistency detection than earlier metrics such as BERTScore or ROUGE. This work established the foundation for applying zero-shot NLI to faithfulness detection in downstream systems, including RAG pipelines, without requiring any task-specific labeling.

A key practical implication of Laban et al.'s findings is that model choice within the NLI framework matters: models trained on large-scale NLI datasets such as MNLI and SNLI exhibit different sensitivity profiles to contradiction, with larger models generally producing more calibrated entailment probabilities. This motivates the comparison between BART-MNLI and DeBERTa-MNLI in the present study.

2.3 Benchmark Datasets for Hallucination Evaluation

Li et al. (2023) introduced HaluEval, a large-scale benchmark specifically designed to evaluate model faithfulness across three task types: question answering, dialogue, and summarization. For each task type, HaluEval contains pairs of (document, generated response) annotated as either hallucinated or faithful. The hallucinated examples were generated by prompting ChatGPT to produce plausible-sounding but factually incorrect responses, and both positive and negative examples were verified by human annotators.

HaluEval is particularly relevant to RAG faithfulness evaluation because the question-answering subset mirrors the RAG output format: a retrieved passage serves as the knowledge source and the generated response is judged against it. The benchmark includes approximately 35,000 annotated QA examples, making it suitable for supervised fine-tuning as well as for zero-shot evaluation. Li et al. also evaluated several LLMs on HaluEval and found that even large models fail to identify a substantial proportion of hallucinated responses, suggesting that this remains an open and practically significant problem.

This study uses HaluEval's QA subset as both the training corpus for the fine-tuning method and the primary evaluation set for all methods. The FEVER fact-verification dataset (Thorne et al., 2018) was originally planned as an out-of-domain generalization test but was not completed in the current experimental cycle.

3. Methodology

The study evaluates multiple methods for detecting whether a generated answer is faithful to a source passage. For all methods, the detection task is treated as a binary classification: an input pair (passage, generated answer) is labeled as faithful or hallucinated. All experiments use a 10,000-example evaluation subset of the HaluEval QA dataset, containing 5,010 positive ("yes" / faithful) and 4,990 negative ("no" / hallucinated) examples. For zero-shot methods, the full 10,000-example set is used as the test set. For supervised fine-tuning, two data-split configurations were tested: an 80/20 train/test split and a 70/15/15 train/validation/test split.

3.1 Method 1 — NLI Zero-Shot Inference

The first method applies pre-trained NLI models directly to the faithfulness detection task without any task-specific training. The rationale is that contradiction detection, which is one of the three standard NLI labels (entailment, neutral, contradiction), corresponds closely to faithfulness failure. An NLI model that assigns high probability to contradiction given the (passage, answer) pair is therefore a natural detector of hallucination.

Two models are used in the standard zero-shot setting. The first is facebook/bart-large-mnli, a BART-large model fine-tuned on the Multi-Genre NLI (MNLI) corpus via zero-shot classification. The second is MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli, a DeBERTa-v3-Large model trained on a combination of MNLI, FEVER, ANLI, LingNLI, and WANLI datasets. DeBERTa-v3 uses disentangled attention over content and position embeddings and applies span-based masking during pre-training, which has been shown to produce stronger natural language understanding representations than BART on NLI benchmarks. The multi-dataset training of this particular checkpoint is expected to provide broader coverage of entailment patterns compared to models trained on MNLI alone.

For each (passage, answer) pair, the NLI model produces probabilities across candidate labels. A prediction of faithful ("yes") is made when the entailment probability exceeds the contradiction probability. The method requires no labeled training data and can be applied directly to any text pair, making it highly portable across domains.

As a variant, we also tested a SummaC-Zero–inspired approach using the BART-MNLI backbone. In this configuration, the passage and answer are decomposed into individual sentences, and NLI scores are computed at the sentence-pair level before being aggregated into a document-level consistency score. This approach follows the methodology of Laban et al. (2021) and is intended to test whether sentence-level decomposition improves detection accuracy compared to full-document NLI inference.

3.2 Method 2 — Supervised Fine-Tuning

The second method fine-tunes a smaller NLI-initialized model on the HaluEval training split. The base model is microsoft/deberta-v3-small, which has approximately 184 million parameters, roughly 5× smaller than DeBERTa-v3-Large. The choice of DeBERTa-v3-small is deliberate: the objective of this method is to produce a model that is both accurate and fast enough for deployment in a latency-sensitive pipeline.

A binary classification head—a linear layer with two output units—is appended to the pre-trained encoder and trained jointly. Training uses the Hugging Face Trainer with default AdamW optimization. The model is trained for 4–5 epochs. Two data-split configurations were tested: the first uses an 80/20 train/test split (8,000 training examples, 2,000 test examples), and the second uses a 70/15/15 train/validation/test split (7,000 training examples, 1,500 validation examples, 1,500 test examples). In both configurations, the label distribution is approximately balanced.

The input format concatenates the passage and the generated answer using the model's standard separator tokens, treating the passage as the premise and the generated answer as the hypothesis. Binary cross-entropy loss is used throughout training. Evaluation metrics including F1, precision, recall, and accuracy are computed at the end of each epoch on the validation or test set.

One recognized limitation of this method is that fine-tuning on HaluEval's distribution may cause the model to overfit to the stylistic patterns of ChatGPT-generated hallucinations. Additionally, as our results demonstrate, the fine-tuning process itself is sensitive to hyperparameter choices and may fail to converge if the learning rate, number of epochs, or input formatting are not carefully calibrated.

3.3 Method 3 — LLM-as-Judge (Planned)

The third method is planned but not yet completed. It uses a large-parameter instruction-following model as an automated evaluator. The evaluator is presented with a structured prompt containing the source passage, the generated answer, and a rubric requesting a binary faithfulness judgment with chain-of-thought rationale. The response is parsed to extract a final label.

The primary purpose of this method is to establish a performance ceiling: given that large models have demonstrated strong instruction-following and reasoning capabilities, the LLM-as-Judge is expected to produce more accurate faithfulness judgments than smaller automated methods, at substantially higher inference cost. Comparing Methods 1 and 2 against Method 3's performance will indicate how much accuracy is sacrificed by using cheaper automated approaches.

Model selection and prompt engineering for Method 3 are deferred to future work.

4. Results

Table 1 summarizes the performance of all methods on the HaluEval test set. The zero-shot methods are evaluated on the full 10,000-example set. Fine-tuning results are reported for the held-out test set under each split configuration.

Method	Accuracy	Precision	Recall	F1	Notes
BART-MNLI (Zero-Shot)	68.8%	0.730	0.598	0.657	No training
DeBERTa-v3-Large-MNLI (Zero-Shot)	68.8%	0.730	0.598	0.657	Multi-NLI trained

BART-MNLI + SummaC-Zero	67.2%	0.709	0.587	0.642	Sentence-level NLI
DeBERTa-v3-small (Fine-Tuned, 80/20)	49.4%	0.494	1.000	0.661	Failed to converge
DeBERTa-v3-small (Fine-Tuned, 70/15/15)	50.1%	0.501	1.000	0.667	Failed to converge
LLM-as-Judge	—	—	—	—	Planned

Table 1. *Performance comparison of all detection methods on the HaluEval QA evaluation set. Fine-tuned models predicted a single class for all examples, indicating training failure. LLM-as-Judge evaluation is planned as future work.*

4.1 Zero-Shot NLI Performance

Both zero-shot NLI models—facebook/bart-large-mnli and MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli—achieved identical performance on the 10,000-example HaluEval test set: 68.8% accuracy, with precision of 0.730, recall of 0.598, and F1-score of 0.657. The prediction distribution for both models was skewed toward the negative class (“no” / hallucinated), with 5,892 examples (58.9%) predicted as “no” versus 4,108 (41.1%) predicted as “yes,” compared to the true distribution of 4,990 “no” and 5,010 “yes.”

The identical results between the two models are notable given their different architectures and training data. BART-MNLI was trained solely on the MNLI corpus, while the DeBERTa variant was trained on five NLI datasets (MNLI, FEVER, ANLI, LingNLI, WANLI). One possible explanation is that both models converge to similar decision boundaries when applied to the zero-shot classification pipeline on this particular dataset, or that the HaluEval examples do not exercise the additional NLI patterns that distinguish multi-dataset training from single-dataset training. Further investigation with different thresholding strategies or prompt formulations would be needed to differentiate these models.

The per-class breakdown shows that both models perform better on the negative class (precision 0.66, recall 0.78, F1 0.71) than on the positive class (precision 0.73, recall 0.60, F1 0.66). This asymmetry suggests that the NLI models are more conservative in labeling responses as faithful, which could be advantageous in high-stakes applications where false negatives (missed hallucinations) are more costly than false positives.

4.2 SummaC-Zero Variant

The SummaC-Zero variant, which applies BART-MNLI at the sentence level and aggregates scores, achieved slightly lower performance than the document-level zero-shot approaches: 67.2% accuracy, precision of 0.709, recall of 0.587, and F1-score of 0.642. This represents a 1.5 percentage-point drop in accuracy and a 0.015 drop in F1 compared to the standard zero-shot baseline.

The slight degradation may be attributed to the characteristics of the HaluEval QA dataset. HaluEval’s generated answers are typically short—often a single sentence or a few sentences—which limits the benefit of sentence-level decomposition. The SummaC approach was originally designed for longer abstractive summaries where individual sentence-level verification can catch localized inconsistencies that document-level NLI might miss. On short answers, the decomposition adds noise from the aggregation step without providing meaningful additional granularity.

4.3 Fine-Tuning Failure Analysis

The fine-tuned DeBERTa-v3-small model failed to learn the hallucination detection task under both data-split configurations. In the 80/20 split (8,000 training examples, 2,000 test), the model achieved 49.4% accuracy with an F1 of 0.661; in the 70/15/15 split (7,000/1,500/1,500), it achieved 50.1% accuracy with F1 of 0.667. In both cases, the model predicted class 1 (“yes” / faithful) for 100% of test examples, yielding perfect recall (1.000) but precision equal to the class’s base rate (approximately 0.50).

The training logs confirm that the model did not converge. Training loss remained near 0.693 throughout all epochs—this value corresponds to $-\ln(0.5)$, the cross-entropy loss of a model that assigns equal probability to both classes for every example. Validation metrics oscillated between two degenerate states: predicting all examples as one class (yielding ~0.50 accuracy and F1 ~0.66) or predicting all examples as the other class (yielding ~0.50 accuracy and F1 = 0.00). This oscillation pattern is characteristic of a model that has not found a gradient signal strong enough to escape a degenerate minimum.

Several factors likely contributed to this failure. First, the learning rate may have been too high or too low for the DeBERTa-v3-small architecture on this task, preventing the classification head from learning meaningful features. Second, the input formatting—concatenating the full passage and answer as a single NLI pair—may exceed the model’s effective context window or dilute the signal that distinguishes faithful from hallucinated answers. Third, DeBERTa-v3-small has 184 million parameters, substantially fewer than the large models used in zero-shot evaluation; this capacity gap may be insufficient for the fine-grained semantic distinctions required by the HaluEval benchmark. Finally, the number of training epochs (4–5) may have been insufficient given the slow convergence, or the optimizer may have required a warmup schedule that was not configured.

5. Discussion

5.1 Zero-Shot NLI as a Practical Baseline

The zero-shot NLI results establish that pre-trained entailment models can detect hallucinations in RAG outputs at approximately 69% accuracy without any task-specific training or labeled data. While this accuracy level is insufficient for fully automated deployment in high-stakes settings, it provides a meaningful signal that can be used for flagging, filtering, or routing responses for human review. The precision of 0.73 on the positive class indicates that when the model classifies a response as faithful, it is correct roughly three-quarters of the time; however, the recall of 0.60 means that approximately 40% of faithful responses are incorrectly flagged as hallucinated.

From a deployment perspective, the zero-shot approach has significant practical advantages. It requires no labeled training data, no fine-tuning infrastructure, and can be applied immediately to any domain by using off-the-shelf NLI models from Hugging Face. For organizations that need a “quick start” hallucination detection capability, zero-shot NLI represents the most accessible entry point.

The identical performance of BART-MNLI and the multi-dataset DeBERTa model raises questions about whether the zero-shot classification pipeline itself is a bottleneck. Both models are applied through Hugging Face’s zero-shot-classification pipeline, which frames the task as entailment between the input text and candidate label descriptions. If the label descriptions (e.g., “yes” and “no”) are insufficiently informative, the pipeline may collapse model differences. Experimenting with more descriptive label prompts—such as “The answer is fully supported by the document” versus “The answer contains information not found in the document”—could potentially differentiate the models and improve accuracy.

5.2 Diagnosing the Fine-Tuning Failure

The failure of supervised fine-tuning is the most instructive result in this study. The degenerate behavior—predicting a single class for all inputs—is a well-known failure mode in classification tasks and typically indicates one or more of the following issues: a learning rate that is too large (causing the model to overshoot useful minima) or too small (preventing meaningful updates), label encoding errors where the mapping between text labels and integer class indices is inconsistent between training and evaluation, a loss function that is not properly connected to the classification head, or input sequences that are truncated in a way that removes the discriminative content.

In the HaluEval context, the input format requires careful handling. Each example contains a question, a knowledge passage, and a generated answer, which together can produce long sequences. If the tokenizer truncates these inputs to a maximum length that cuts off the answer portion, the model effectively sees only the passage and must predict faithfulness without access to the text it is supposed to evaluate. This truncation hypothesis is consistent with the degenerate behavior observed and should be the first issue investigated in future work.

Additionally, the HaluEval dataset presents a subtle challenge for NLI-based fine-tuning: the hallucinated examples were generated by ChatGPT and are designed to be plausible near-copies of faithful answers with localized factual substitutions. Distinguishing these requires the model to perform precise token-level comparison rather than the broader semantic matching that NLI models are typically trained for. A smaller model like DeBERTa-v3-small may lack sufficient capacity for this level of fine-grained discrimination, particularly when initialized from an NLI checkpoint that has been optimized for a different task distribution.

5.3 Implications for Practitioners

The practical takeaway from these results is twofold. First, zero-shot NLI provides a usable—if imperfect—hallucination detection signal that can be deployed immediately with no additional training cost. For applications where approximate detection is acceptable (e.g., flagging suspicious responses for human review), this approach is viable. Second, fine-tuning a hallucination detector is not as straightforward as applying a standard text classification recipe. Practitioners should expect to invest effort in hyperparameter search, input formatting validation, and potentially architectural modifications before achieving competitive fine-tuned performance.

The SummaC-Zero variant's slight underperformance relative to document-level NLI suggests that for short-form QA outputs, sentence decomposition does not add value and may introduce noise. This finding may not generalize to longer-form outputs such as multi-paragraph summaries, where sentence-level NLI has been shown to be more effective in prior work (Laban et al., 2021).

5.4 Limitations

Several limitations constrain the conclusions that can be drawn from the current results. First, the evaluation was conducted on a 10,000-example subset of HaluEval rather than the full 35,000-example dataset, which may affect the generalizability of the reported metrics. Second, the fine-tuning experiments used a limited hyperparameter search; a more systematic grid search or learning-rate schedule exploration would be necessary to determine whether the convergence failure is fundamental or configuration-dependent. Third, the FEVER cross-dataset evaluation was not completed, leaving the question of cross-domain generalization unanswered. Fourth, all experiments were conducted using the Hugging Face zero-shot-classification pipeline, which imposes its own prompt formatting; alternative inference strategies might yield different results. Finally, the LLM-as-Judge method was not completed, so the performance ceiling for automated detection on this dataset remains unknown.

5.5 Future Work

The immediate next steps include diagnosing and resolving the fine-tuning convergence failure through systematic hyperparameter search (learning rate sweep from 1e-6 to 5e-5, warmup ratio variation, sequence length analysis), completing the FEVER cross-dataset evaluation for the zero-shot methods, and implementing the LLM-as-Judge baseline. Beyond these, experimenting with alternative label descriptions in the zero-shot pipeline, testing prompt-based classification with instruction-tuned models, and fine-tuning larger DeBERTa variants (base or large) are promising directions for improving detection accuracy.

6. Conclusion

This study evaluated multiple methods for detecting faithfulness failures in RAG-generated outputs using the HaluEval QA benchmark. Zero-shot NLI using both facebook/bart-large-mnli and MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli achieved 68.8% accuracy and F1-scores of 0.657, establishing a practical baseline that requires no task-specific training data. A SummaC-Zero sentence-level aggregation variant slightly underperformed the document-level approach at 67.2% accuracy (F1 = 0.642), suggesting that sentence decomposition does not benefit short-form QA evaluation.

Supervised fine-tuning of DeBERTa-v3-small on HaluEval failed to converge across two split configurations, with the model consistently predicting a single class and achieving approximately chance-level accuracy (~50%). This negative result highlights that hallucination detection via fine-tuning is not a straightforward classification task and requires careful attention to hyperparameters, input formatting, and model capacity. The training failure itself constitutes a useful finding: it demonstrates that smaller models with standard fine-tuning recipes cannot simply be plugged into hallucination detection without substantial configuration effort.

The central finding is that zero-shot NLI provides the most accessible and immediately deployable approach to hallucination detection for RAG systems, achieving moderate accuracy with zero training overhead. For practitioners seeking to deploy hallucination detection in production, zero-shot NLI represents a strong starting point, with fine-tuning offering potential improvements only when the training pipeline is carefully validated. The LLM-as-Judge evaluation, planned as future work, will establish the accuracy ceiling for automated detection on this benchmark.

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